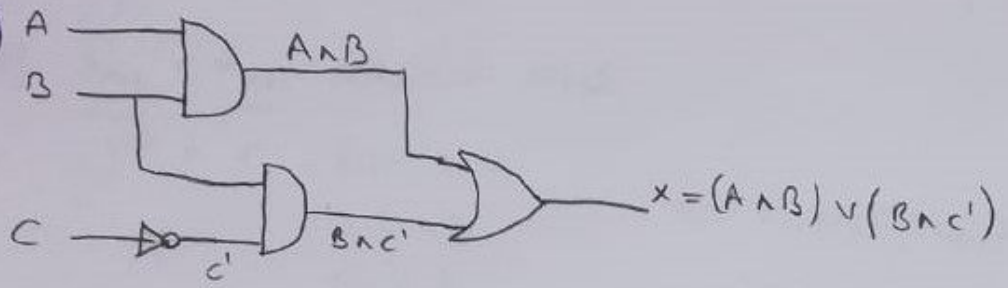
"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally on my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

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150718017

Q1

2)



b)

A	B	A∧B
0	0	0
0	1	0
1	0	0
1	1	1

B	C	B∨C
0	0	0
0	1	1
1	0	1
1	1	1

A	B	C	C'	A∧B	B∧C'	x=(A∧B) v (B∧C')
1	1	1	0	1	0	1

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q_2

a) $a_{n+2} + a_{n+1} - 6a_n = 0 \quad n \geq 0$

$r^2 + r - 6r = 0$

$r_1 = -3$

$r_2 = 2$

$a_n = c_1 \cdot (-3)^n + c_2 \cdot 2^n$

$a_0 = -2/c_1 + c_2 = 0$

$a_1 = -3c_1 + 2c_2 = 5$

$-5c_1 = 5$

$c_1 = -1 \quad c_2 = 1$

$a_n = (-1)(-3)^n + (1) \cdot 2^n$

b) $a_{n+2} + 8a_{n+1} - 9a_n = 0$

$r^2 + r^2 - 9 = 0$

$r_1 = -9$

$r_2 = 1$

$a_n = c_1 \cdot (-9)^n + c_2 \cdot 1^n$

$a_0 = c_1 + c_2 = 0$

$a_1 = -9c_1 + c_2 = 10$

$10c_1 = -10$

$c_1 = -1$

$c_2 = +1$

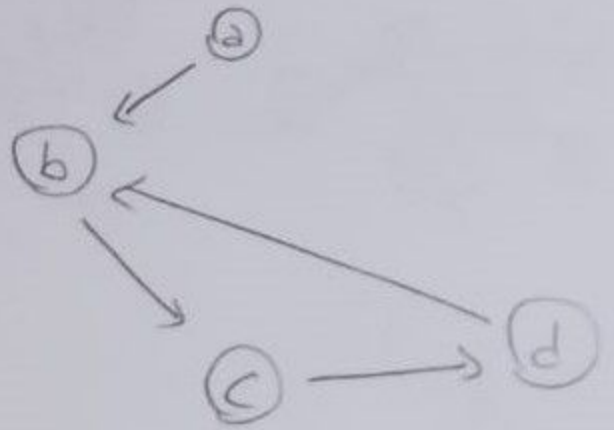
$a_n = (-1) \cdot (-9)^n + (1) \cdot 1^n$

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150718017
Ads

Q3

a)



b) R^* of R =

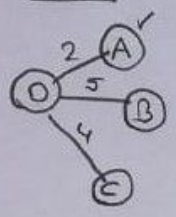
$\{(a,b),(a,c),(b,c),(b,d),(c,b),(c,d),(d,b),(d,c)\}$

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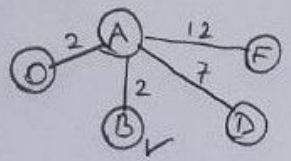
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9 5 7

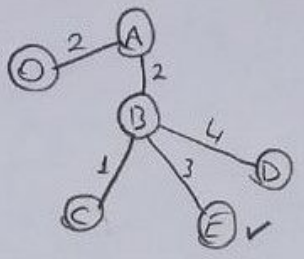
Step 1



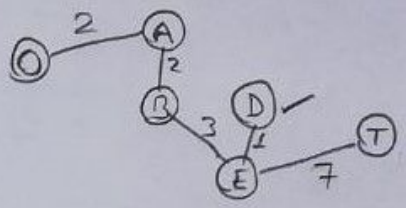
Step 2



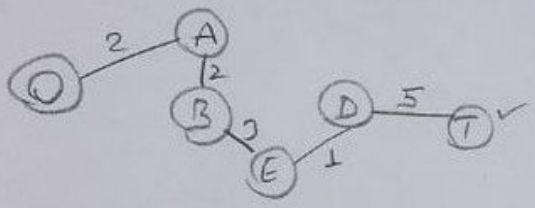
Step 3



Step 4



Step 5

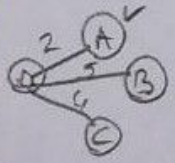


$2+2+3+1+5 = 13$

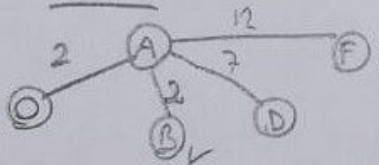
$O-A-B-E-D-T$

↳ This is one of the shortest path

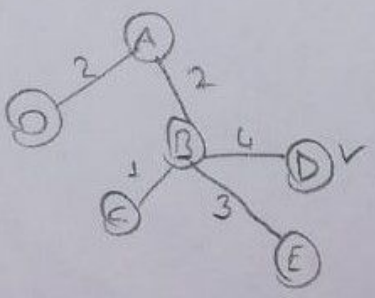
Step 1



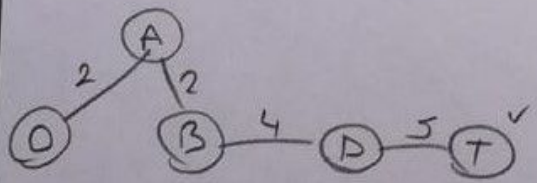
Step 2



Step 3



Step 4



$2+2+4+5 = 13$

$O-A-B-D-T$

↳ This is the another one of the shortest path